

Reading Revit (`.rvt`) Files Without Revit

Test Report

RNGD side-testing project

September 23, 2026

1 Summary

Three archives were extracted from `Testtt/`, yielding three Revit 2022 sample models. **The architecture and structural archives contain the byte-identical file** (`racbasicsampleproject.rvt`, same MD5), so only two distinct models exist. Using only `olefile` and `zlib` we can read: file metadata, the embedded thumbnail, and the element database's text (level, room, family and view names). We *cannot* recover 3D geometry; that needs Revit, Autodesk Platform Services or Dynamo. The included tools are a Python module (`rvt_tools.py`), a CLI, a Streamlit viewer, and a 17-test pytest suite (all passing).

2 Method

An `.rvt` is an OLE2 compound file. `BasicFileInfo` gives the Revit version and build; `RevitPreview4.0` holds a PNG thumbnail; `Partitions/12` is the element database, stored as many raw-deflate chunks behind gzip headers. Decompressing every valid chunk and scanning for UTF-16 strings yields names of levels, rooms and families. Category counts are keyword hit counts, a heuristic and not exact element counts.

3 Architecture model

File	<code>racbasicsampleproject.rvt</code>
Size	18.61 MB
Revit version / build	2022 / 20210129_1515(x64)



Figure 1: Embedded preview thumbnail (128 px original, upscaled 3×).

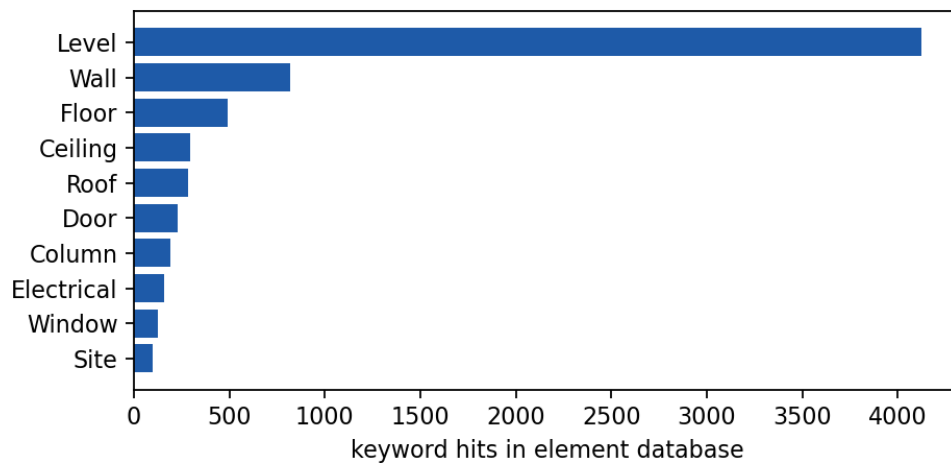


Figure 2: Category keyword signal (heuristic).

Sample recovered names: levels: Level Head - Upgrade, Level 1, Level 2, Floor Plan Level 1; bedrooms: Bedroom, Master Bedroom, Dormitory Bedroom); kitchen items: Vault(TM) offset kitchen sink with four-hole faucet drilling, 4500_Kitchen Island, Kitchen & Dining.

4 Mep model

File	rmeadvancedsampleproject.rvt
Size	36.61 MB
Revit version / build	2022 / 20210129_1515(x64)



Figure 3: Embedded preview thumbnail (128 px original, upscaled $3\times$).

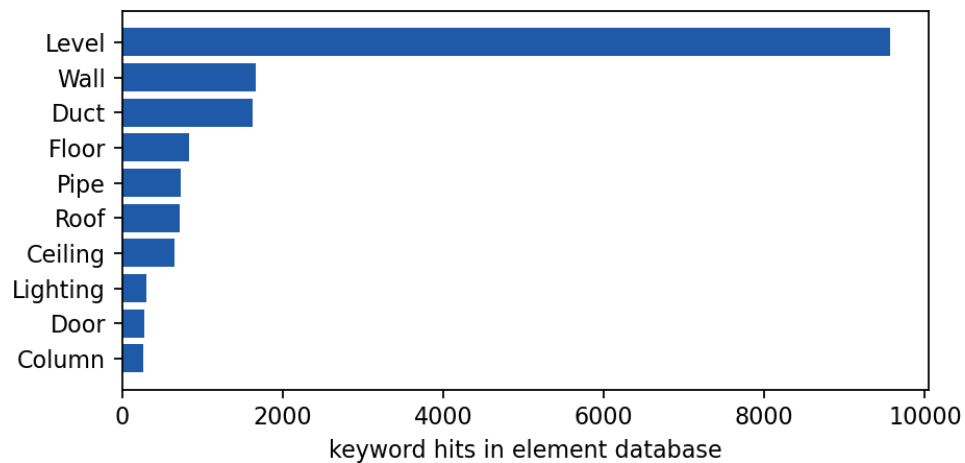


Figure 4: Category keyword signal (heuristic).

Sample recovered names: levels: Level 1, Level Head - Upgrade, Level 2, Level 3; bedrooms: none; kitchen items: Sinks - Kitchen.

5 Structural model

File	racbasicsampleproject.rvt
Size	18.61 MB
Revit version / build	2022 / 20210129_1515(x64)



Figure 5: Embedded preview thumbnail (128 px original, upscaled 3×).

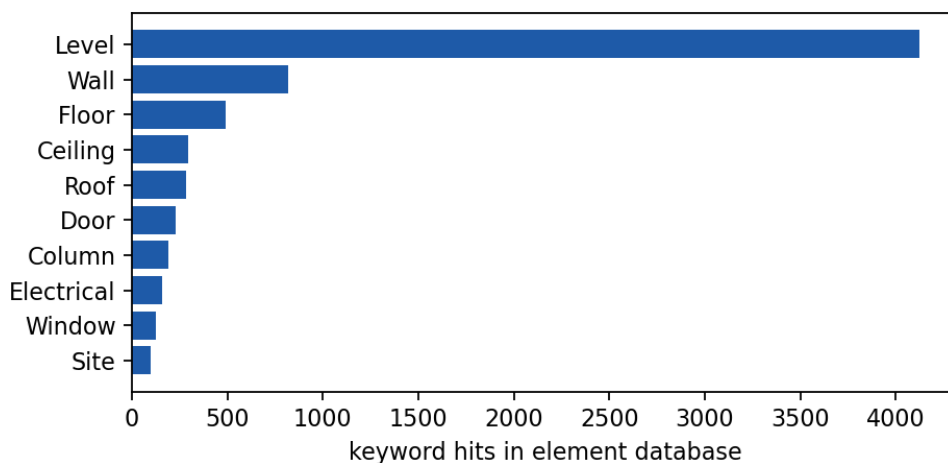


Figure 6: Category keyword signal (heuristic).

Sample recovered names: levels: Level Head - Upgrade, Level 1, Level 2, Floor Plan Level 1; bedrooms: Bedroom, Master Bedroom, Dormitory Bedroom); kitchen items: Vault(TM) offset kitchen sink with four-hole faucet drilling, 4500_Kitchen Island, Kitchen & Dining.

6 pyRevit integration

pyRevit v6.5.5.26237 was installed per-user (%APPDATA%/pyRevit-Master). **Revit itself is not installed on this machine** (pyrevit env lists no Revit installs), so pyRevit cannot attach and scripts that use the Revit API cannot run here. Two layers were built:

1. **Works now, no Revit:** pyrevit_tools.py wraps pyrevit revits fileinfo, which reads

project name, client, build, workshared flag and document GUID directly from the file. Results (identical files de-duplicated):

File	Project name	Client	Doc ID	Workshared
racbasicsampleproject.rvt	Sample House	Autodesk	c26966e9	No
rmeadvancedsampleproject.rvt	Project Name	Owner	74792680	No

2. **Needs Revit:** the extension `pyrevit_extension/RNGDTest.extension` (registered in pyRevit's search path) adds an *RNGD* ribbon tab with two buttons: *Summarize model* (element counts per category, levels, rooms with areas, wall types → JSON) and *Export walls* (center-lines, heights, types → JSON). These are **untested** until Revit is available; `pyrevit_tools.load_revit_sum` reads their output.

The RVT sample files were saved by Revit 2022 (build 20210129); opening them needs Revit 2022 or newer.

7 Limitations and next steps

- Only a 128 px thumbnail is embedded; there is no geometry to render here.
- For real geometry/quantities: Autodesk Platform Services Model Derivative API (cloud, needs an APS key), or Revit/Dynamo/pyRevit locally.
- The string harvest is heuristic; treat category counts as relative signal only.
- Relevance to RNGD: these sample models could seed the BaaP catalog work once extraction via APS or Revit is available.